

Supplementary Material: Theory of Augmentation for Improved Prediction Accuracy: Evaluating the Predictive Utility of Synthetic Liver Cirrhosis Data with Consensus-Based Quality Filtering

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1 Scope of This Document

The main manuscript reports the headline results within a strict page budget, which means a good deal of supporting detail had to be left out. This document supplies that detail. It is organised so that each section can be read on its own. All sixteen figures are cited directly from the main paper as Fig. S1 to Fig. S16. The eight tables, numbered Table S1 to Table S8, contain material that does not appear in the main paper at all.

Three kinds of material appear here. The first is methodological detail that a reader would need in order to reimplement the pipeline exactly: full architecture and optimiser settings, the preprocessing decisions, and the seeding regime that makes the results reproducible. The second is the complete per-feature statistical output. The main paper quotes a handful of Kolmogorov-Smirnov statistics to illustrate a pattern; Tables S2 to S4 give all of them, for every feature and every generative method, so that the pattern can be checked rather than taken on trust. The third is the set of sensitivity and ablation analyses that justify particular parameter choices, chiefly the consensus tolerance of 5.0 and the output-perturbation level of $\sigma = 0.7$.

The figures and distributional tables describe the saved full-training artifacts, not newly trained fold-specific pools. The main manuscript now reports a separate corrected five-fold generator rerun, with per-fold provenance and outputs included in the revision package. Historical supplementary plots must not be mistaken for that rerun.

2 Extended Methodology

Fig. S16 gives an overview of the end-to-end pipeline referenced throughout this document and the main paper.

2.1 Dataset and Preprocessing

The Mayo Clinic Primary Biliary Cirrhosis trial data contains 418 patient records across 20 columns. Patients 1 to 312 were enrolled in the randomised placebo-controlled trial of D-penicillamine; patients 313 to 418 form a later observational cohort who consented to have basic measurements recorded but who did not undergo the full laboratory panel. Each of those 106 records is missing between nine and ten laboratory values.

This matters for how the missingness is handled. Imputing nine or ten values per record would mean inventing the majority of each patient's biochemical profile, and doing so from a cohort measured under a different protocol. The

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Pearson Correlation Matrix — Continuous Clinical Features (276 complete cases)

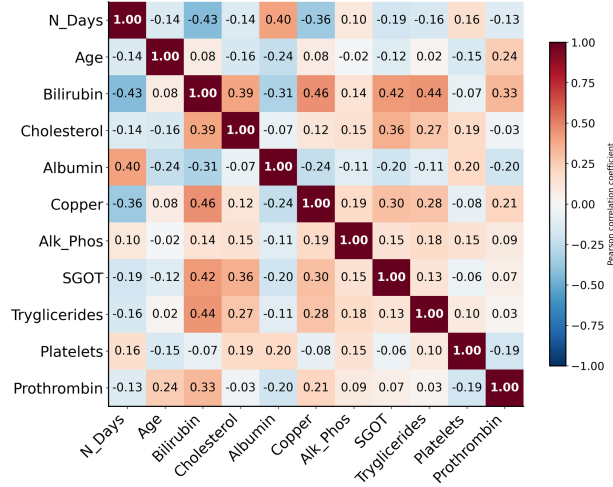


Fig. S1. Pearson correlation matrix across all 276 complete cases. Bilirubin correlates positively with Copper ($r = 0.46$), SGOT ($r = 0.42$), Cholesterol ($r = 0.39$) and Prothrombin ($r = 0.33$), reflecting the interlinked nature of hepatic synthetic and excretory function in PBC. N_Days correlates negatively with Bilirubin ($r = -0.43$) and Copper ($r = -0.36$), consistent with more severe biochemical profiles being associated with shorter survival.

imputed values would carry the statistical structure of the trial arm rather than of the observational arm, which is precisely the kind of systematic bias that would then propagate into the generative models. Complete-case analysis was therefore chosen, leaving 276 records. The cost is a reduction in sample size and a restriction of the analysis to the randomised population; the benefit is that every value used is a value that was actually measured.

The patient identifier column is discarded as it carries no clinical signal. The remaining 19 columns comprise 11 continuous laboratory measurements, five binary clinical indicators, two ordinal variables, and the binary survival outcome. Categorical encodings are given in Section 3 of the main paper.

For the full-training generative analysis, min-max scaling is fitted to the 193 real training patients and inverted after generation. The synthetic table has 19 columns: 18 generated variables and Status. Classifiers exclude N_Days and use 17 baseline predictors, with separate training-fitted standardisation. In the corrected cross-validation, every scaler, generator, IQR bound, and consensus pool is fitted or constructed anew inside the corresponding training fold. Fig. S1 shows the correlation structure of the cohort the generative models are asked to reproduce. Bilirubin sits at the centre of a cluster of correlated markers of hepatic function, and N_Days runs against them, which is the structure any useful synthetic record must preserve.

2.2 Generative Architectures

Table S1 gives the complete specification of all three models. All are implemented directly in TensorFlow 2.15 without third-party synthetic-data libraries, which means every architectural choice is visible and auditable rather than inherited from a package default.

Both the cGAN and the VAE are named for what they are rather than after published models they resemble, and the distinction matters when comparing these results against the literature.

Table S1. Complete Architecture and Optimiser Settings for the Three Generative Models

Setting	Vanilla GAN	cGAN	VAE
Latent dimension	100	100 + 2 (one-hot)	32
Encoder / discriminator	256, 128	256, 128	128, 64
Generator / decoder	128, 256	128, 256	64, 128
Hidden activation	LeakyReLU (0.2)	LeakyReLU (0.2)	ReLU
Output activation	Sigmoid	Sigmoid	Sigmoid
Output dimension	19	19	19
Batch normalisation	Yes (0.8)	Yes (0.8)	No
Dropout	0.3 (disc.)	0.3 (disc.)	No
Optimiser	Adam	Adam	Adam
Learning rate	2×10^{-4}	2×10^{-4}	1×10^{-3}
β_1	0.5	0.5	default
Batch size	32	32	32
Epochs	500	300	300
Conditioning	None	Outcome label	None
Loss	Adversarial	Adversarial	ELBO ($\beta = 1.0$)
Records generated	500	500	500

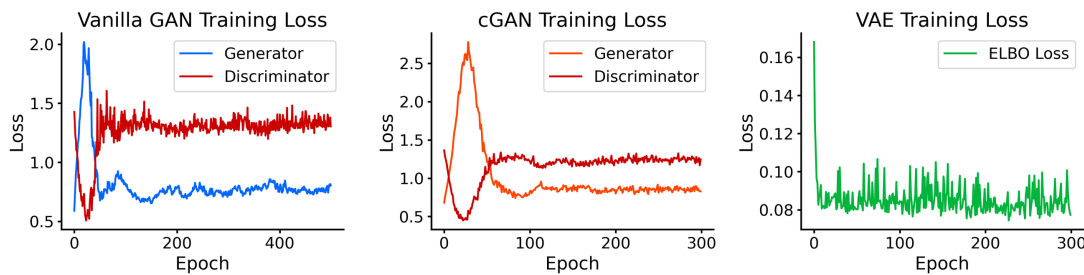


Fig. S2. Saved full-training loss curves for the three core generative models. They describe optimisation, not demonstrated convergence, and are not loss curves from the corrected cross-validation rerun.

The model labelled cGAN is a conditional GAN in the sense of Mirza and Osindero [1]: it conditions generator and discriminator on the outcome label and nothing more. It is deliberately not called CTGAN, because it implements none of the three components that define the Conditional Tabular GAN of Xu et al. [2], namely mode-specific normalisation, training-by-sampling, and a PacGAN critic with the Wasserstein gradient penalty. A full CTGAN was implemented separately and compared against it; that comparison is reported in Section 4 of the main paper, where CTGAN reaches FID 0.108 against 0.072 for the cGAN after 4,000 generator updates, with convergence not established.

The model labelled VAE is likewise a standard variational autoencoder applied to min-max scaled tabular data. It is not the TVAE of Xu et al. [2], which shares that paper's mode-specific normalisation, and results reported here should not be read as a benchmark of TVAE as originally specified.

Training curves for all three models are shown in Fig. S2. These curves describe optimisation behaviour but do not establish convergence or exclude mode collapse.

Table S2. Two-Sample Kolmogorov-Smirnov Statistics Against Real Training Data. Lower is Better. An Asterisk Marks a Statistically Significant Difference at $\alpha = 0.05$

Feature	GAN	cGAN	VAE	Consensus
N_Days	0.194*	0.075	0.396*	0.209*
Age	0.083	0.122	0.410*	0.214*
Bilirubin	0.413*	0.487*	0.485*	0.269*
Cholesterol	0.308*	0.454*	0.341*	0.172*
Albumin	0.315*	0.079	0.389*	0.194*
Copper	0.262*	0.286*	0.420*	0.174*
Alk_Phos	0.506*	0.409*	0.423*	0.249*
SGOT	0.165*	0.072	0.425*	0.171*
Triglycerides	0.081	0.179*	0.301*	0.122*
Platelets	0.149*	0.148*	0.405*	0.190*
Prothrombin	0.092	0.116	0.352*	0.133*

2.3 Reproducibility

The full-training artifact analyses use seed 42. The corrected five-fold generator rerun uses seeds 42–46, with split and classifier seeds fixed at 42. Its environment includes Python 3.9.6, TensorFlow 2.15.0, NumPy 1.26.4, and scikit-learn 1.5.2. The run completed 15 generator fits at the original epoch settings. Fold membership, scaling extrema, IQR bounds, synthetic records, and individual fold AUCs are retained. Fixed seeds alone do not ensure identical results across software versions or hardware.

Two practical notes for anyone reproducing this work. First, the consensus procedure used throughout is the *equalised* variant described in Section 3 of the main paper, in which the VAE pool is downsampled to match the smallest adversarial pool before voting. An earlier non-equalised variant produces a differently sized consensus set and different downstream numbers. Second, the nearest-neighbour privacy analysis reads the datasets written by `master_runner.py` rather than regenerating them, so it always describes the same synthetic records the rest of the results are computed from.

3 Complete Distributional Fidelity Results

The main paper summarises distributional fidelity with a single aggregate FID per method and quotes a few representative Kolmogorov-Smirnov statistics. That summary hides an important structure, which the full tables make visible: the methods do not fail uniformly across features, and the method with the best aggregate score is not the best on every feature.

3.1 Kolmogorov-Smirnov Tests

Table S2 gives the two-sample KS statistic between each synthetic pool and the real training data, for all 11 continuous features.

Three observations follow. The VAE differs significantly from the real data on all 11 features, which is consistent with its poor aggregate FID of 0.2280 and reflects the over-smoothing characteristic of variational reconstruction. The adversarial models each pass on a handful of features and fail badly on the heavily right-skewed ones: both GAN and cGAN produce their worst statistics on Bilirubin, Cholesterol and Alkaline Phosphatase, the three most skewed

Table S3. Jensen-Shannon Divergence Between Synthetic and Real Training Distributions. Lower is Better

Feature	GAN	cGAN	VAE	Consensus
N_Days	0.070	0.039	0.401	0.098
Age	0.056	0.064	0.397	0.110
Bilirubin	0.072	0.077	0.248	0.057
Cholesterol	0.084	0.150	0.140	0.057
Albumin	0.091	0.040	0.304	0.065
Copper	0.063	0.069	0.250	0.071
Alk_Phos	0.138	0.110	0.205	0.080
SGOT	0.033	0.030	0.325	0.062
Triglycerides	0.021	0.048	0.179	0.039
Platelets	0.045	0.055	0.356	0.095
Prothrombin	0.023	0.030	0.219	0.047

biomarkers in the cohort. This is the expected failure mode, since a sigmoid output layer trained on min-max scaled data has difficulty reproducing a long right tail.

The consensus set behaves differently from either of its parents. It is significant on all 11 features, so by the binary reading of the test it looks worse than the individual adversarial pools. But its *magnitudes* are the most uniform of any method, and on the hardest features it is substantially the best: Bilirubin 0.269 against the GAN's 0.413 and cGAN's 0.487, and Alkaline Phosphatase 0.249 against 0.506 and 0.409. Cross-model agreement trades a small loss on the easy features for a large gain on the difficult ones. This is worth stating carefully, because it is the clearest evidence that consensus voting does what it was designed to do at the distributional level, even though Section 4 of the main paper shows it delivers no predictive benefit.

3.2 Jensen-Shannon Divergence

Table S3 reports the Jensen-Shannon divergence per feature, which is bounded in $[0, 1]$ and, unlike the KS statistic, is sensitive to differences across the whole distribution rather than at the point of maximum separation.

The divergence measure tells a cleaner story than the KS test. The VAE is between three and ten times worse than every other method on nearly every feature, with N_Days and Age the worst affected at roughly 0.40. The consensus set is the most consistent performer: it never exceeds 0.110 on any feature, whereas the GAN reaches 0.138 on Alkaline Phosphatase and cGAN reaches 0.150 on Cholesterol. Averaged across features the consensus set is the best of the four, which is the result the aggregate FID of 0.0809 also reflects.

3.3 Standardised Mean Differences

Table S4 gives Cohen's d between each synthetic pool and the real training data. A positive value means the synthetic mean exceeds the real mean.

The VAE has small absolute standardised mean differences on all 11 variables (maximum 0.230), but it is not the closest method for every variable. For example, its Albumin and SGOT differences exceed those of cGAN. Small mean differences should not be interpreted as agreement of the entire distribution.

One possible interpretation is that concentration around central values preserves means while reducing variability. This is not a demonstrated mechanism or an inevitable property of VAEs. The KS and Jensen-Shannon results reveal distributional differences not captured by standardised mean differences.

Table S4. Cohen’s d Between Synthetic and Real Training Data. Values Near Zero Indicate Well-Matched Means

Feature	GAN	cGAN	VAE	Consensus
N_Days	−0.245	+0.100	+0.099	−0.025
Age	−0.012	+0.007	−0.072	+0.076
Bilirubin	+0.686	+0.695	−0.016	+0.530
Cholesterol	+0.581	+0.661	−0.044	+0.426
Albumin	−0.644	+0.059	+0.163	−0.193
Copper	+0.607	+0.546	−0.104	+0.462
Alk_Phos	+0.771	+0.667	−0.025	+0.531
SGOT	+0.362	+0.089	−0.230	+0.108
Triglycerides	+0.229	+0.303	+0.017	+0.281
Platelets	+0.248	+0.175	−0.088	+0.067
Prothrombin	+0.159	+0.227	−0.045	+0.197

The results illustrate that quality rankings depend on the selected criterion. Mean agreement, distributional-shape agreement, and predictive usefulness are different properties and should be measured separately. This dataset does not establish a universal ranking of generative models.

3.4 Supporting Figures

Fig. S13 shows the IQR-based Tukey fence filter (Section 3 of the main paper) in action on the Bilirubin distribution, the hardest continuous feature to reproduce faithfully.

Four further figures make the tables above visible. Fig. S3 plots the marginal distributions of six representative biomarkers side by side for all four synthetic datasets. It shows directly what the KS statistics in Table S2 quantify: the adversarial models track the real distribution closely on Albumin and Prothrombin, while every method struggles on the heavily right-skewed Bilirubin, and the VAE distribution is displaced rather than merely broadened.

Fig. S4 moves from single features to their joint structure, comparing the real correlation matrix against the consensus set. The dominant clinical associations survive generation, but the weaker off-diagonal terms are attenuated, which is the expected signature of a generator that has learned the strongest dependencies and smoothed the rest.

Figure S5 provides an exploratory two-dimensional projection of real and synthetic records. Apparent overlap or concentration in this projection does not establish high-dimensional fidelity, a causal explanation for privacy risk, or record-level disclosure. Those questions require separate quantitative analyses.

4 Consensus Threshold Sensitivity

The consensus tolerance of 5.0 standardised units is the single most consequential free parameter in the pipeline, so it was selected by a systematic sweep rather than by inspection. Table S5 reports the sweep across eight values.

The class-balance column is the reason a very tight tolerance is unusable. At 2.5 the accepted set is 1.1% deceased, against 40.4% in the real training cohort. A strict agreement requirement does not sample the outcome classes evenly: deceased patients occupy the sparser, more extreme region of biomarker space, exactly where three independently trained models are least likely to place records near one another. Tightening the tolerance therefore filters out the minority class almost entirely, and augmenting with such a set would worsen the imbalance the augmentation was meant to relieve.

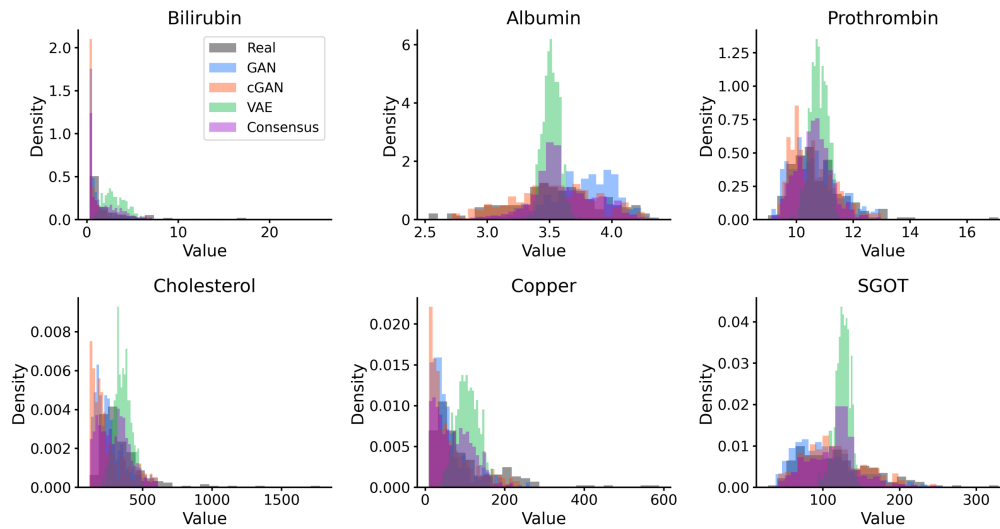


Fig. S3. Marginal distributions of six representative biomarkers for real training data and all four synthetic datasets. GAN and cGAN reproduce Albumin and Prothrombin closely. All methods diverge on the heavily right-skewed Bilirubin and Cholesterol, and the VAE distribution for Bilirubin is shifted noticeably rightward, matching the KS statistics in Table S2.

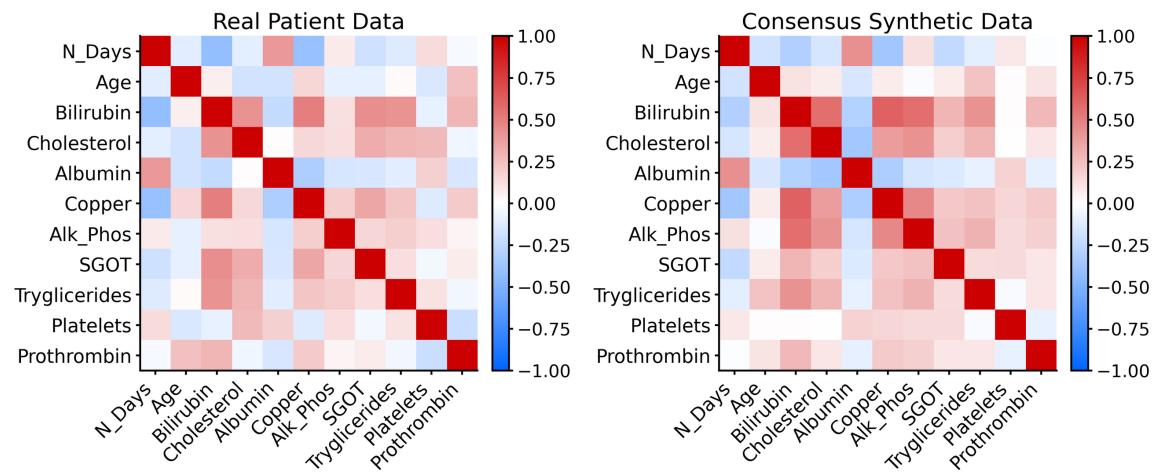


Fig. S4. Pearson correlation structure of real training data (left) against the equalised consensus set (right). Dominant clinical associations are preserved, including the positive Bilirubin-Copper and negative N_Days-Bilirubin relationships, while weaker correlations are attenuated.

As the tolerance widens, class balance recovers monotonically and FID improves monotonically, but the marginal return falls away. Between 5.0 and 6.0 the accepted set grows by 90 records while FID improves by only 0.0121, and each additional record is by construction one that fewer models agreed on. The value of 5.0 sits at the point where class balance has become tolerable at 24.7% and the quality curve has begun to flatten. It is a defensible compromise rather than an optimum, and we state it as such.

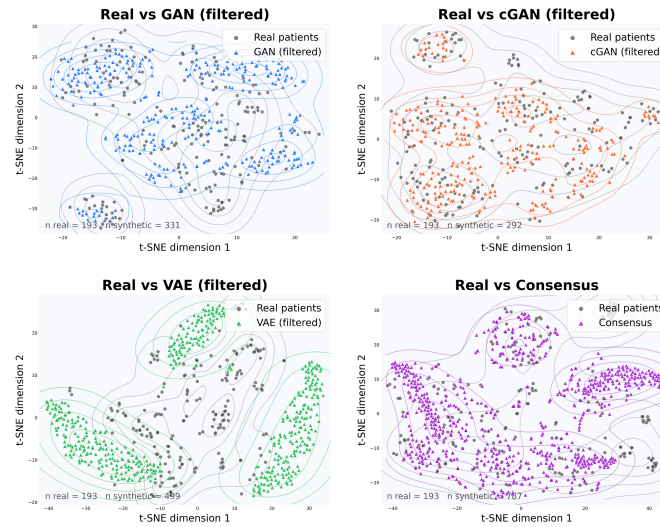


Fig. S5. t-SNE projection of real patients and each synthetic method from the original 18-dimensional feature space. Filtered GAN and cGAN records overlap the real manifold. VAE records spread beyond the real density contours. The consensus set concentrates in the central high-density region, but the projection does not establish a causal explanation for the separately measured 21.1% near-duplicate rate.

Table S5. Consensus Tolerance Sweep. Composition Percentages Are Shares of the Accepted Set. The Real Training Cohort Is 40.4% Deceased

Tol.	Records	GAN %	cGAN %	VAE %	Deceased %	FID
2.5	89	11.2	15.7	73.0	1.1	0.2262
3.0	252	20.6	15.1	64.3	10.7	0.1677
3.5	461	21.5	16.1	62.5	17.4	0.1441
4.0	667	22.6	17.4	60.0	21.1	0.1247
4.5	848	23.5	20.8	55.8	23.2	0.1061
5.0	953	25.1	22.6	52.4	24.7	0.0950
5.5	1017	27.4	23.5	49.1	25.8	0.0864
6.0	1043	28.7	23.5	47.8	26.3	0.0829

Note that the record counts in this table are from the non-equalised voting procedure, which is why 5.0 yields 953 records here against the 787 reported in the main paper. The equalisation step, which downsamples the VAE pool before voting, is applied afterwards and reduces the count while improving fidelity.

5 Extended Privacy Analyses

Fig. S14 gives the full nearest-neighbour privacy risk analysis referenced in Section 4 of the main paper (near-duplicate rates in its disclosure-risk table).

Table S6. Gaussian Output Perturbation Applied to the Filtered VAE Pool ($n = 499$)

σ	Near-dup. (n)	Near-dup. (%)	FID
0.0	178	35.7	0.2280
0.1	169	33.9	0.2095
0.2	140	28.1	0.1720
0.3	88	17.6	0.1402
0.5	29	5.8	0.0804
0.7	7	1.4	0.0481
1.0	2	0.4	0.0319

Table S7. Approximate DP-SGD Accounting Projections (Not Certified) at $\delta = 10^{-5}$ for $n = 193$ Training Patients

Noise multiplier	ϵ	Interpretation
0.5	210.4	Unvalidated projection
0.8	89.2	Unvalidated projection
1.0	61.2	Unvalidated projection
1.2	46.0	Unvalidated projection
1.5	33.6	Unvalidated projection
2.0	23.9	Unvalidated projection
3.0	14.0	Unvalidated projection

5.1 Output Perturbation

The filtered VAE pool has a 35.7% near-duplicate rate under the specified distance threshold. This is an empirical proximity indicator, not a probability of re-identification. Table S6 reports post-hoc Gaussian perturbation of the eleven continuous variables, with noise measured in training-standard-deviation units.

Both columns improve together, which is not what one would expect. Adding noise to synthetic records normally buys privacy at the cost of fidelity; here FID improves from 0.2280 to 0.0481 while the near-duplicate rate falls from 35.7% to 1.4%.

The lower FID after perturbation is consistent with a change in the moments of the synthetic distribution, but it does not establish the mechanism responsible or demonstrate improved downstream utility. Noise can also produce implausible clinical combinations, so plausibility and predictive performance need separate reassessment.

The $\sigma = 0.7$ setting is reported as an illustrative point in the sweep. A near-duplicate rate of 1.4% is not a validated release criterion or formal disclosure guarantee.

5.2 Illustrative Privacy-Accounting Projections

Table S7 reports approximate analytical projections from a simplified DP-SGD accounting formula at $\delta = 10^{-5}$ for 193 training records. No differentially private generator was trained for this sweep. The approximation is not a validated accountant and the numbers must not be interpreted as certified privacy budgets.

The projections depend on the assumed sampling and composition settings. They do not establish a minimum cohort size for differential privacy, and there is no universal numerical threshold separating meaningful from meaningless protection. A valid guarantee requires a correctly implemented private training mechanism and validated accounting.

Table S8. Subgroup Test-Set AUC, N_Days Excluded, Reported as Scenario A / Scenario C. Subgroups Below $n = 20$ Are Reported for Completeness Only

Subgroup	n	Dec.	RF	GB	LR
Early stage (1–2)	17	4	0.587 / 0.596	0.615 / 0.596	0.635 / 0.654
Late stage (3–4)	66	29	0.848 / 0.852	0.857 / 0.859	0.863 / 0.859
Female	72	27	0.839 / 0.835	0.843 / 0.831	0.844 / 0.834
Male	11	6	0.517 / 0.567	0.533 / 0.633	0.600 / 0.767
D-penicillamine	39	17	0.765 / 0.763	0.778 / 0.775	0.818 / 0.856
Placebo	44	16	0.884 / 0.891	0.877 / 0.864	0.853 / 0.842

Output perturbation is an empirical proximity experiment and must be distinguished from differentially private training. Neither the perturbation results nor these projections certify the release of the present synthetic records.

Both analyses in this section are plotted together in Fig. S15.

6 Subgroup Analysis

Table S8 reports test-set AUC within six clinical strata, comparing the real-only baseline against consensus augmentation, on the leak-free 17-feature input (N_Days excluded). Each cell gives Scenario A followed by Scenario C.

Among the four subgroups large enough to support inference, the picture is consistent with the overall null result. Differences between Scenario A and Scenario C stay within about ± 0.04 , with no consistent direction, and every AUC exceeds 0.76. Augmentation neither helps nor harms in any stratum we can measure reliably.

The two small subgroups illustrate why we decline to interpret them. On the 11 male patients, all three classifiers move in the same direction this time (Random Forest 0.517 to 0.567, Gradient Boosting 0.533 to 0.633, Logistic Regression 0.600 to 0.767), which on six deceased patients is as consistent with a single reordered prediction shifting AUC by several points as with a real effect; the subgroup is too small to distinguish the two. The early-stage subgroup, with 17 patients and four deceased, is equally unstable and additionally sits near chance for all three classifiers.

We report these rows rather than dropping them because selective omission of inconvenient strata is itself a reporting problem. But no weight should be placed on them, and a reader who wishes to cite a subgroup result should use the late-stage, female, or treatment-arm rows.

7 Graphical Views of the Predictive Results

These plots display the saved held-out classification results, which were reproduced at their reported precision in the artifact audit. They are not cross-validation plots; the corrected fold-contained results are reported separately in the main manuscript.

Figure S6 compares held-out precision–recall curves. Their interpretation depends on the event prevalence in the evaluation cohort, not only training-set balance. Visual proximity of curves does not demonstrate equivalence or establish a null effect.

Figure S7 compares held-out AUC across Scenarios A–C. All displayed AUCs exceed 0.80, but this threshold does not establish clinical adequacy or prove that augmentation cannot harm a model. The changes differ by classifier and should be interpreted with their uncertainty.

Fig. S8 shows F1 as a heatmap, which makes the classifier-specific pattern easier to see than a bar chart does. Gradient Boosting is the only classifier whose F1 declines monotonically as synthetic volume increases from Scenario A through

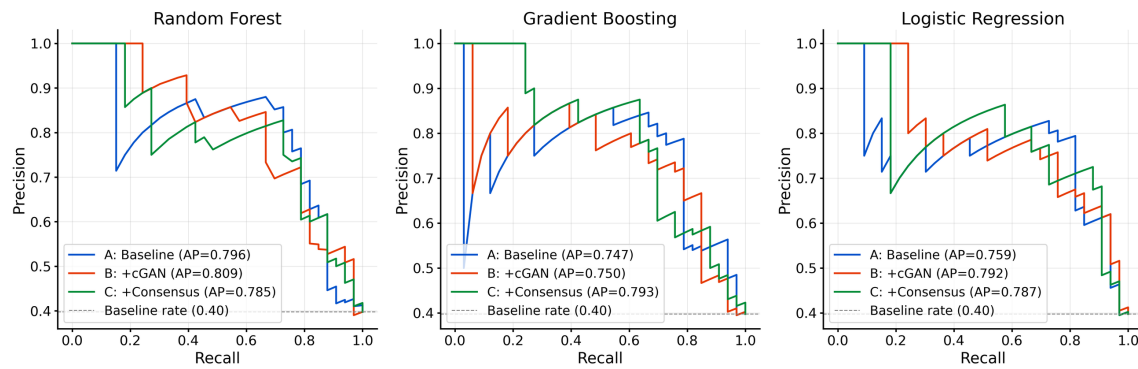


Fig. S6. Precision-Recall curves by classifier and training scenario on the 83-patient real held-out test set. PR curves are informative for this cohort given its class imbalance of 78 deceased against 115 survivors in the training set. All classifiers hold precision above 0.7 across a wide recall range.

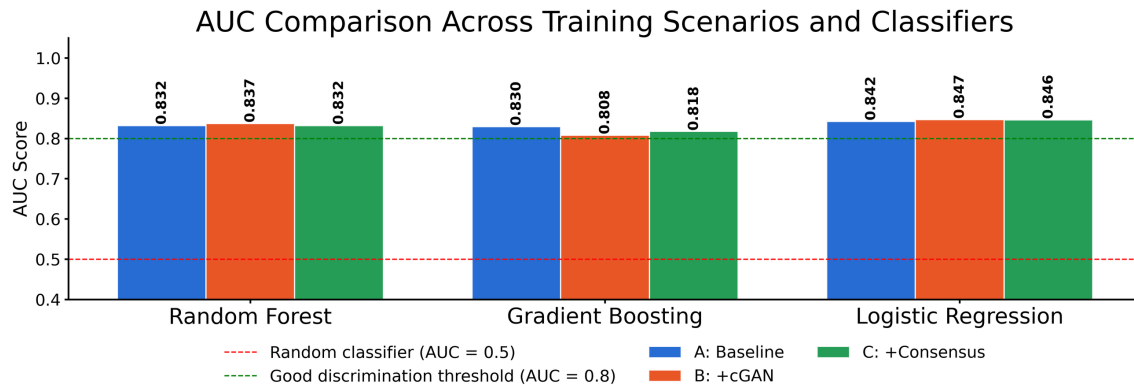


Fig. S7. AUC across all three classifiers for scenarios A, B and C, N_Days excluded from classifier input. The green dashed line marks AUC = 0.8 and the red dashed line marks AUC = 0.5. Every combination clears 0.8. Logistic Regression under Scenario B achieves the highest AUC at 0.847.

Scenario C. This is consistent with the mechanism argued in the main paper: a boosting model fitted on a sufficient real signal has least to gain from additional records drawn from the dense centre of the distribution, and most to lose from the shift in class weighting they introduce.

Figs. S9 and S10 present the same results grouped two further ways, the first by metric across all five scenarios and the second as metric trajectories per classifier. Both make the flatness of the comparison plain. Scenario E, trained on synthetic data alone, is the only condition that separates clearly from the rest, and it does so downwards for Gradient Boosting while remaining competitive for Random Forest and Logistic Regression.

The grouped view in Fig. S9 holds accuracy, F1 and AUC side by side so the three metrics can be compared within a scenario. The trajectory view that follows takes the opposite cut, following each metric across scenarios for a single classifier at a time.

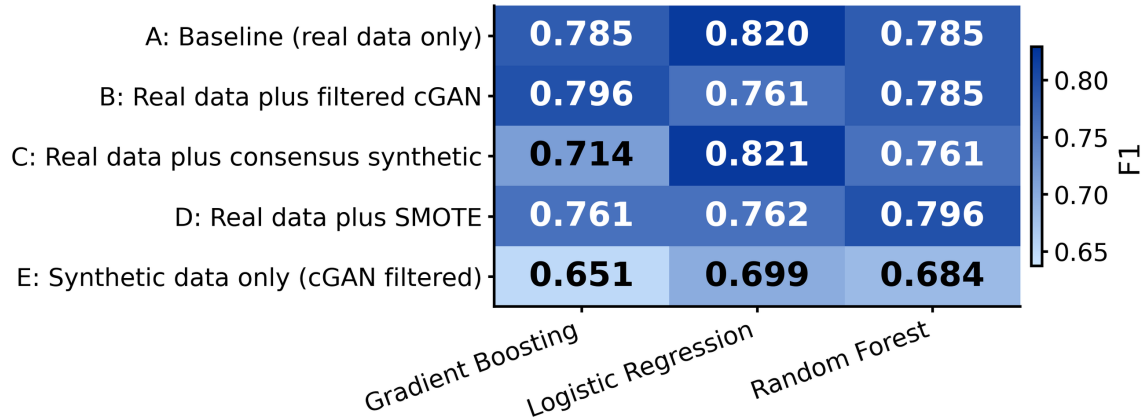


Fig. S8. F1-score heatmap across training scenarios, N_{Days} excluded from classifier input; darker shading indicates higher F1. Logistic Regression under Scenario C achieves the highest F1 at 0.821. Under Scenario C, Random Forest falls to 0.761 and Gradient Boosting to 0.714 while Logistic Regression is essentially unchanged (0.820 to 0.821), showing that consensus augmentation does not move all classifiers in the same direction.

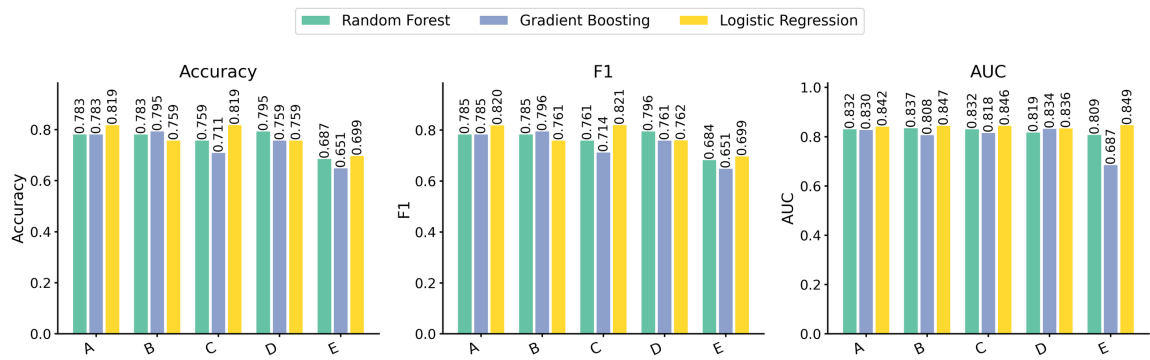


Fig. S9. Grouped comparison of accuracy, F1 and AUC across all training scenarios and classifiers. No augmentation scenario uniformly outperforms the real-data baseline.

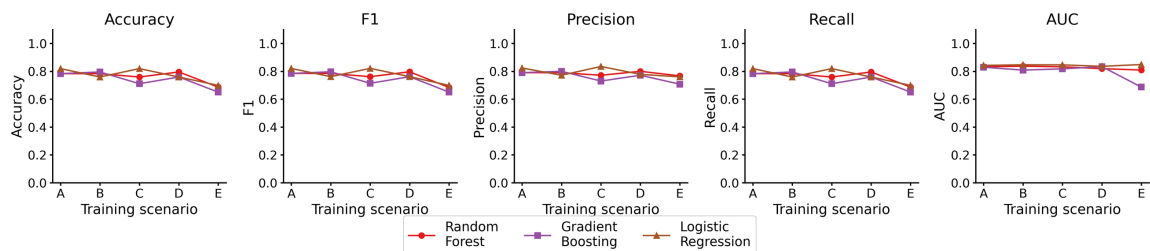


Fig. S10. All five evaluation metrics across training scenarios for each classifier, N_{Days} excluded from classifier input. Trajectories are flat or slightly declining from Scenario A through Scenarios B and C, with Scenario D (SMOTE) producing a modest accuracy gain for Random Forest only (0.783 to 0.795); Gradient Boosting and Logistic Regression both decline under Scenario D.

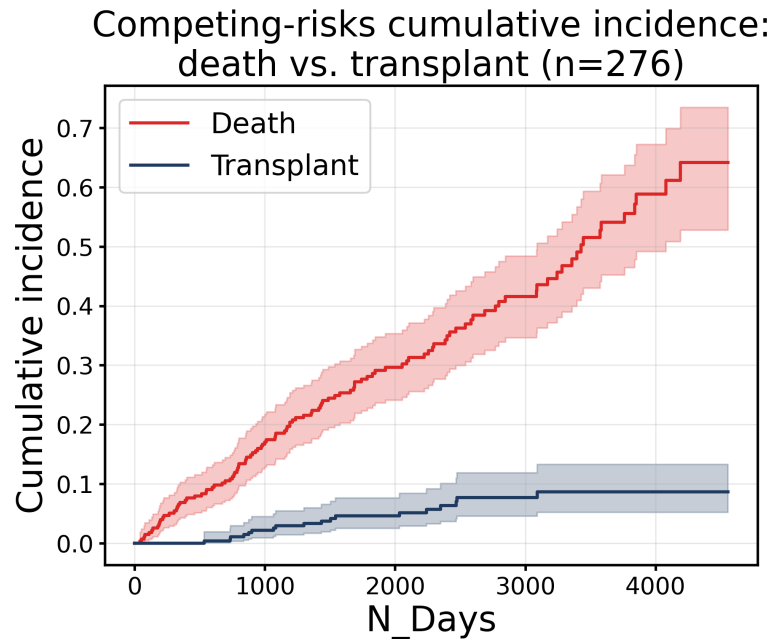


Fig. S11. Aalen-Johansen cumulative incidence of death and transplantation as competing events, computed on all 276 complete-case patients (shaded bands are 95% confidence intervals). Death accounts for the large majority of observed events (64.2% cumulative incidence by end of follow-up versus 8.7% for transplant), describing the observed competing-event burden without establishing that censoring transplantation has negligible consequences for prediction.

8 Competing Risks: Death Versus Transplant

The survival-modelling framing in Section 4 of the main paper (Cox proportional hazards, Random Survival Forest) treats death as the event of interest and censors at transplantation, alongside patients alive at data cutoff. This is a cause-specific hazard for death, a standard and defensible choice when death is the endpoint of interest, but it assumes transplant timing carries no information about death risk beyond what the covariates capture, an assumption worth checking rather than asserting, since transplant priority is itself driven by disease severity.

Of 276 complete cases, 111 died, 18 received a transplant, and 147 were censored alive. Figure S11 displays Aalen-Johansen cumulative incidence with transplantation treated as a competing event. Estimated end-of-follow-up incidence is 64.2% for death and 8.7% for transplantation. This estimator accounts for competing events but still requires appropriate assumptions about other right censoring. The transplant frequency alone does not establish negligible effects on risk prediction. Cause-specific hazards and cumulative incidence answer different questions; a competing-risks regression was not fitted.

9 Kaplan-Meier Comparison Across Six Generators

Section 4 of the main paper reports log-rank test statistics comparing each generator's synthetic (N_Days , $Status$) distribution against the real training cohort, as a second check on whether FID predicts downstream utility. Fig. S12 gives the underlying Kaplan-Meier curves for all six generators.

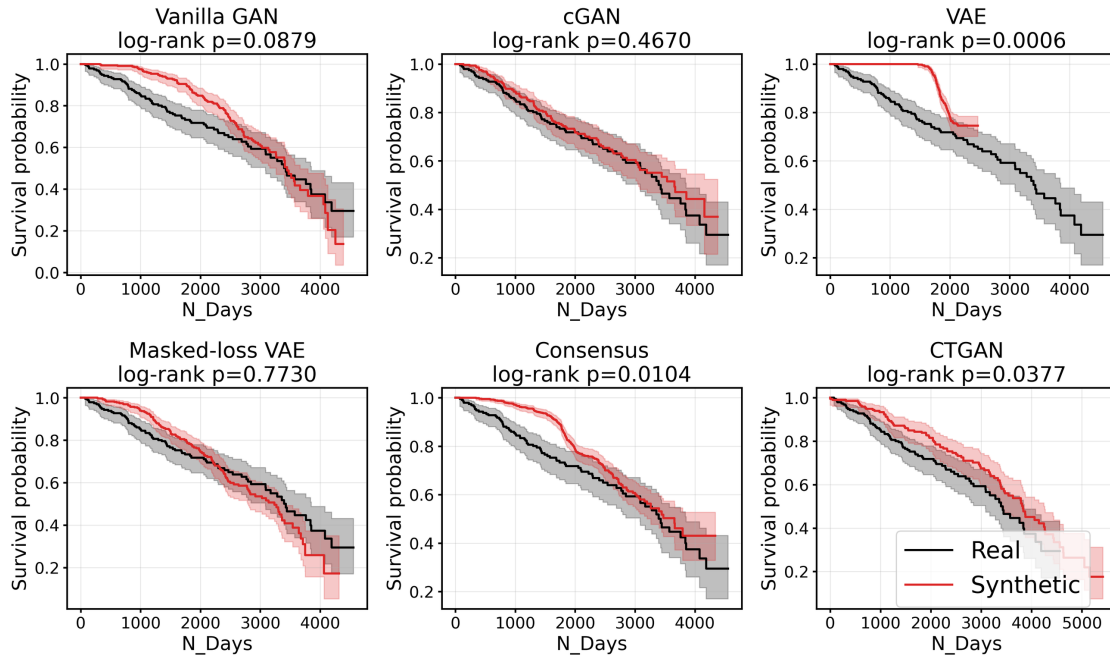


Fig. S12. Kaplan-Meier survival curves for each generator's synthetic (N_Days, Status) pool (red) against the real training cohort (black), with log-rank test p -values. The comparisons are exploratory, unadjusted for multiplicity, and dependent on the reference cohort used for generation. Non-significant tests do not establish equivalent survival distributions.

10 Data and Code Availability

The Mayo Clinic PBC dataset is publicly available from the UCI Machine Learning Repository as "Cirrhosis Patient Survival Prediction" (doi: 10.24432/C5R02G), <https://archive.ics.uci.edu/dataset/878/cirrhosis+patient+survival+prediction>. All code required to reproduce every number and figure in the main paper and in this document is available at <https://github.com/Michaeludousoro/liver-cirrhosis-synthetic-data>, including the generative models, the IQR and consensus filters, the evaluation pipeline, and the scripts that produce each supplementary table.

11 Additional Supporting Figures

The four figures below are referenced from earlier sections of this document (Extended Methodology and Extended Privacy Analyses) and are placed together here to keep this document's figure numbering stable as material is added.

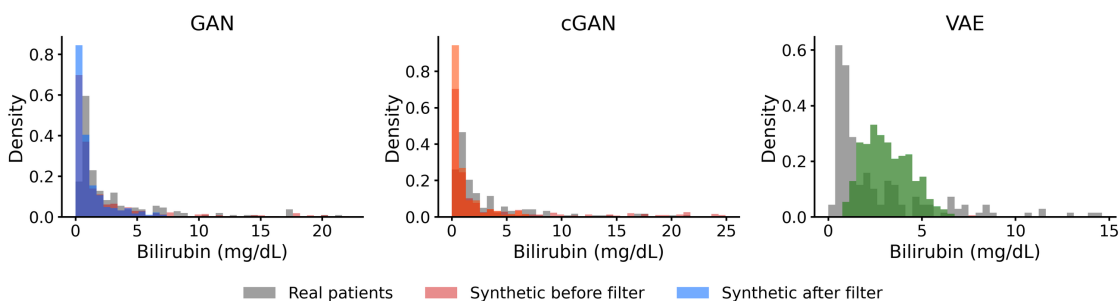


Fig. S13. Effect of IQR-based Tukey fence filtering on the Bilirubin distribution for all three generative models. Grey bars show real patient density. Pink bars show unfiltered synthetic distribution; coloured bars show the post-filter distribution. The GAN and cGAN produce records extending beyond 15 mg/dL that the filter removes. The VAE retains 499 of 500 generated records but produces a distribution shifted rightward relative to real patients.

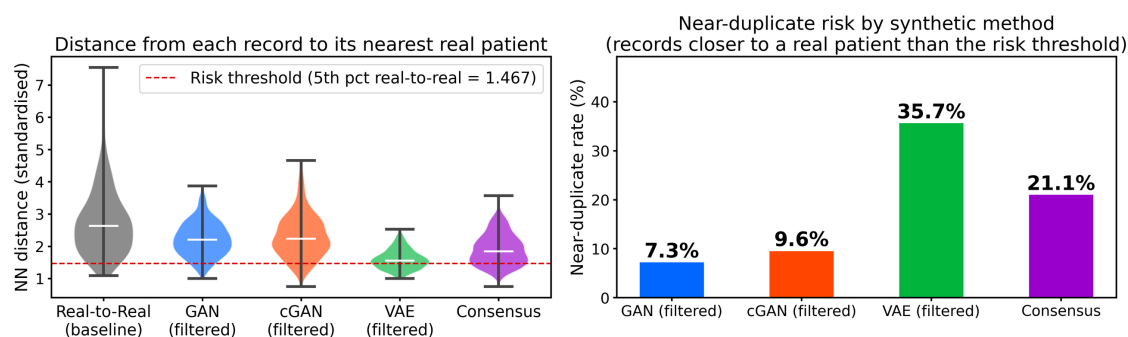


Fig. S14. Privacy disclosure risk analysis using nearest-neighbour distances in standardised feature space. Left: violin plots of distances from each synthetic record to its closest real training patient. The red dashed line marks the risk threshold at the 5th percentile of real-to-real distances. GAN and cGAN distributions are centred above this threshold. Right: near-duplicate rates by method. The observed rates are 7.3% for GAN, 9.6% for cGAN, 35.7% for VAE, and 21.1% for consensus. Lower proximity rates do not certify safe release.

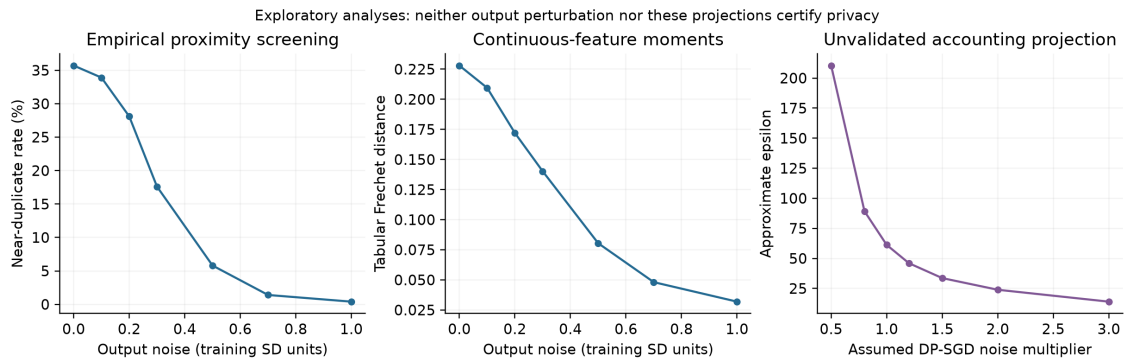


Fig. S15. VAE privacy enhancement analysis. Left: near-duplicate rate versus Gaussian output noise σ (in feature standard-deviation units); at $\sigma = 0.7$ the rate falls from 35.7% to 1.4%, crossing the 5% near-duplicate target. Centre: tabular FID score versus σ ; FID improves from 0.228 to 0.0481, below the equalised consensus benchmark of 0.0809. Right: estimated DP-SGD privacy budget ϵ versus noise multiplier at $\delta = 10^{-5}$; ϵ remains above 14 even at a multiplier of 3.0, under the illustrative approximation; this is not a trained or certified DP model.

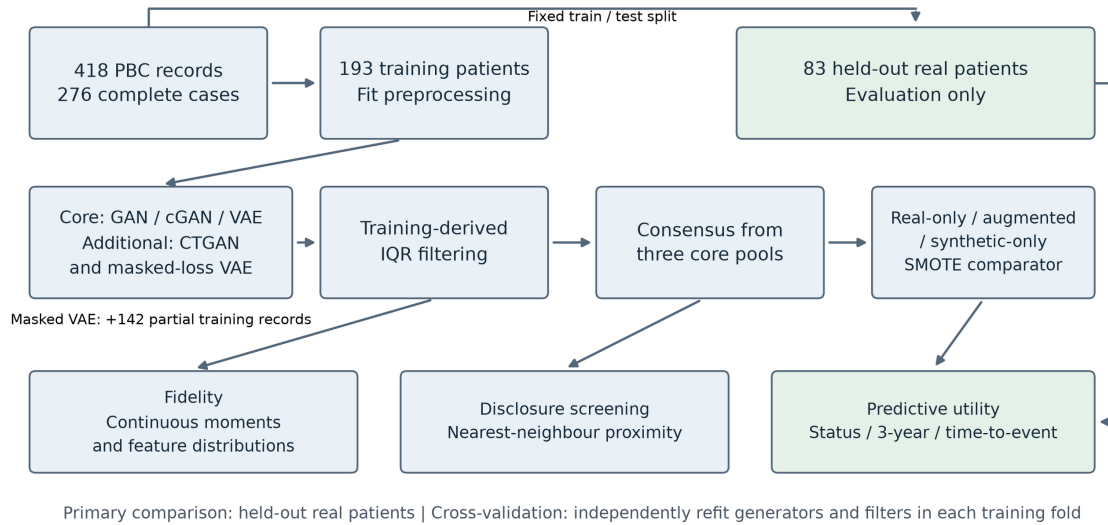


Fig. S16. Updated evaluation pipeline. The 276 complete cases are split into 193 training and 83 held-out patients before fitting pre-processing or generators. The corrected cross-validation independently refits generators, filters, and consensus inside each training fold; the test cohort is used only for evaluation.

References

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